

Worst case sorts

HeapSort = $O(n \lg n)$

MergeSort $O(n \lg n)$

-HeapSort sorts in place, so no additional memory is needed. MergeSort does well if no RAM is available.

-The best sorting algorithm can only do $O(n)$ because one must look at all element

Dynamic Programming

-Divide and conquer doesn't always work

-binary search, merge sort use d&c

-a d&c algorithm for Fibonacci numbers is an exponential algorithm

-Programming - an organized set of activities

-Store results of small subproblems and look them up instead of recomputing

-Usually applied to an optimization problem

-Used for problems where there are many solutions, but looking for the best one

-For shortest path, it is a n^3 algorithm

Dijkstra's

-shortest path first

-used in networking and routing (OSPF)

Longest Common Subsequence

-Two items, find the longest sequence shared between the two (items can be skipped)

-Between abaca and bbabc, the LCS is abc

-Similar to diff in UNIX